English Quantified Partitives are derived by Extraposition plus NP-deletion

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Some English quantifiers may appear in *quantified partitives* (1). But not all may (2).

(1) a. Many/some of the cakes

b. Much/some of the cake

(2) I ate {all/each/*every} of the fish.

There are two main extant analyses of these constructions. One is the 'surface true' analysis, in which the quantifier takes a PP-argument (Lobeck, 1995, Barker, 1998, Gagnon, 2013, Shin, 2016, or a DP with *of*-insertion; Matthewson, 2000, 2001):

(3) [$_{DP}$ some [$_{PP}$ of the cake]]

An alternative analysis takes there to be a silent noun serving as the quantifier's argument (Jackendoff, 1968, Hoeksema, 1984, Kayne, 2007, 2025, Dobrovie-Sorin and Giurgea, 2021, i.a.). The silent noun is often treated as meaning something like 'unit'.

(4) a. Much UNIT of the cake (Jackendoff, 1968)

b. Much AMOUNT of the cake (Kayne, 2007)

c. Many NUMBER of the cakes

The observed distribution of NP-modifiers in quantified partitives poses a problem for both of these theories. I will argue that the best theory includes a duplicated noun which is deleted (see also Zamparelli, 1998, Sauerland and Yatsushiro, 2004). But my proposal differs from previous theories as it is not merely the *head* of the NP which is silent, but the whole NP which is deleted (on silent N vs elided NP, see Merchant, 2023).

(5) a. Many ~~[$_{NP}$ cakes]~~ of the cakes

b. Much ~~[$_{NP}$ cake]~~ of the cake

Phrasal deletion is the crucial ingredient in capturing – and unifying – two generalizations which I will illustrate in Section 2. First, only the quantifiers which may appear without 'bare' – that is, without an overt restrictor – may appear in quantified partitives. Second, only extraposable NP-modifiers may modify quantified partitives.

1 Objections to silent noun and 'surface true' theories of quantified partitives

The objections to both theories of quantified partitives come from the distribution of NP-modifiers. In English, quantified partitives allow prepositional phrases and relative clauses, but not adjectives (for similar observations, see Jackendoff, 1977; Gagnon, 2013).

(6) The bakery produced ten cakes.

a.	Some iced cakes were on the shelf.	(AP; not partitive)
b.	Some of them (with cherries on top) were on the shelf.	(PP; partitive)
c.	Some of them (which were iced by Al) were on the shelf.	(RC; partitive)
d.	Some (*iced) of them were on the shelf.	(*AP; partitive)

In this section, I argue that silent noun theories overgenerate, and that 'surface true' theories undergenerate. In the next section, I turn to a finer-grained characterization of which NP-modifiers can appear, arguing that this motivates the NP-deletion theory.

1.1 Silent noun theories overgenerate

Emphasizing facts like (6d), Gagnon (2013) and Shin (2016) object to a silent noun in quantified partitives, since (7)'s ungrammaticality would have to be stipulated:

(7) *some [$_{\text{NP}}$ iced $\varnothing_{\text{N}}$ of them]

Can English adjectives otherwise modify silent nouns? Perhaps. When they occur bare in plural definite contexts, there is arguably silent *people* (Kayne, 2017, Merchant, 2023):

(8) CONTEXT: at a kennel, looking at dogs.
The very obedient #(ones) will be rewarded with treats.

As the reading obtained with *ones* omitted can only be about very obedient people, not the very obedient dogs, it suggests the existence of silent NP *people*.

Shin also objects that the silent noun theory cannot straightforwardly explain why the silent noun is not available as the restrictor to any quantifier:

(9) *{every/a/the/no/my} $\varnothing_N$ of them

The silent noun theory faces two distinct overgeneration problems. It then seems simpler to allow the quantifiers which can appear in quantified partitives to optionally select partitive PPs, as in (10):

(10) $[_{DP_1}$ all $[_{PP}$ of $[_{DP_2}$ them$]]]$ (Gagnon, 2013, Shin, 2016)

But the proposal in (10) solves the overgeneration problem at the cost of creating an undergeneration problem.

1.2 'Surface true' theories undergenerate

The 'surface true' analysis provides no appropriate adjunction site for relative clauses and prepositional phrases, such as those in the sentences below:

(11) Al iced some of the cakes. *All of them with vanilla frosting* are beautiful.

(12) *All of them that had white frosting* looked delicious.

These are restrictive modifiers of the head noun, and not appositives: (12) means that all the cakes that had white frosting looked delicious, not that all the cakes looked delicious, and they have white frosting.

Furthermore, polarity items are licensed in the relative clause by negative quantifiers:

(13) None of them [that Al (ever) put (any) decorations on] looked less than perfect.

Appositives, on the other hand, do not allow for NPIs, as shown in (14) (assuming proper names can be modified only by appositive relative clauses; Jackendoff, 1977).

(14) a. At the party, no one greeted men who (ever) wore (any) colorful socks.

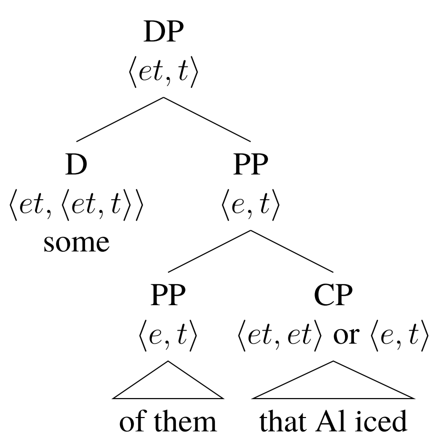
b. At the party, no one greeted John, who (*ever) wore (*any) colorful socks.

The modifiers in (11)-(13) are not appositive, then. What are they modifying? The hypothesis in (10) gives us only three options: the quantificational DP, DP_1; the partitive PP; and the partitioned DP, DP_2. I will now show that none of these are appropriate adjunction sites, so the syntactic structure must be richer than in (10).

In many English dialects, pronouns allow modification, as in *he who drank from my coffee cup*, but only with a 'grandiose' feeling (Elbourne, 2005). This is absent in the above constructions. Some English dialects allow for relative clauses and PPs to modify pronouns directly (allowing *them with vanilla frosting*). But the constructions in (11)-(13) are possible even in dialects (such as mine) that do not allow **them with vanilla frosting*. So the relative clause does not modify the partitioned DP.

If the relative clause were adjoined not to the partitioned DP but to the partitive PP restricting the quantifier, the semantic composition would at least proceed smoothly.

(15) (to be rejected)



However, this hypothesis predicts that [of them that Al iced] is a PP-constituent. The partitive-PP cannot function as if it is modified by the relative clause (see Jackendoff, 1977 and 2013 for a similar observation):

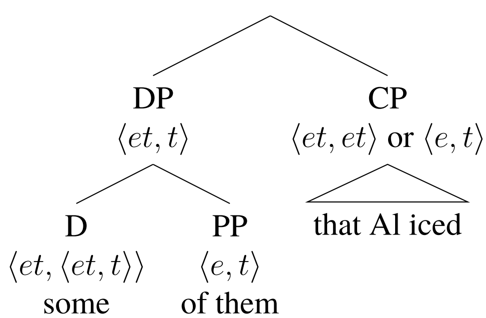
(16) a. Of them, [the group that you met] aren't here anymore.

b. *Of them that you met, [the group t] aren't here anymore.

If (15) was the right constituent structure, fronting would be required to target the lower PP node while being *unable* to target the higher one. Such a rule is unformulable.

This leaves only one possibility – that the relative clause is adjoined high, to the quantificational DP, DP_1. However, under conventional assumptions, the semantic composition fails if the relative clause is adjoined there:

(17) (to be rejected)



If the quantifier could instead be type $\langle et, \langle et, \langle et, t\rangle\rangle\rangle$, this would not be a problem (Bach and Cooper, 1978). Should we want to maintain generalized quantifier theory where determiners take exactly two arguments (Barwise and Cooper, 1981), it is a problem.

Summarizing, the syntax of quantified partitives proposed in Gagnon 2013 and Shin 2016 makes three positions for NP-modifiers available in principle. None of these work, either for syntactic or semantic reasons. But NP-modifiers are attested in quantified partitives, so, these accounts undergenerate. There must be an additional silent adjunction-site for NP-modifiers.

I will now turn to bare quantifiers to establish an independent generalization: only extraposable modifiers are permitted in these. I will then build on the observation that the same quantifiers are permitted bare as are allowed in the partitive. I show how an NP-deletion theory can account for all of these facts.

2 Only the possibly-bare quantifiers and the extraposable modifiers are allowed

2.1 The bareness generalization

The quantifiers which can appear in quantified partitives are also those which can appear 'bare', i.e., without an overt restrictor (Hoeksema, 1984, Kayne, 2007, Shin, 2016), as shown in (18a). If they cannot appear with the partitive, they cannot appear bare (18b).

(18) a. I fed {all, both, each, few, most, none, several, some, three...} (of the sheep)
b. *I fed {a, every, the, no} (of the sheep)

I call this the bareness generalization.

In (18a), the NP (or partitive PP) restricting the quantifiers is only optionally realized, despite the fact that there is no general optionality of NP realization in English:

(19) a. *No linguist likes syntax, but every likes phonology.
b. Good linguists like phonology but bad *(ones) don't. (adapted, Kayne, 2007, ex. 183)

So, if there can be a silent NP restricting the quantifiers in (18a), its licensing is lexically conditioned by that set of quantifiers (Lobeck, 1995). Recall that the silent noun theory fails to derive the bareness generalization (i.e., block (9) without stipulation):

(9) *{every/a/the/no/my} $\varnothing_N$ of them

A theory of quantified partitives should not render it coincidence that the same quantifiers which may appear in partitives may appear bare (as the silent N theory does).

2.2 Only extraposable modifiers are allowed with bare quantifiers; this is explained by NP-deletion

Bare quantifiers (dis)allow the same modifiers as the partitives (compare (20) to (6)):

(20) a. Some (with cherries on top) were on the shelf. (PP)

b. Some (which were iced by Al) were on the shelf. (RC)

c. Some (*iced) were on the shelf. (*AP)

It is not merely leftward/rightward status that explains whether a modifier is available with bare quantifiers/quantified partitives (cf. Jackendoff, 1971). Other rightward modifiers like *a*-adjectives and reduced relative clauses are unavailable with bare quantifiers:

(21) The sleep study we are running is going well. Look at the participants.

a. {The ones/*some} asleep will be woken up soon.

b. Some ??(that) the experimenters examined showed interesting data.

In this, bare quantifiers differ from 'indefinite' quantifiers, which allow adjectives (Abney, 1987, Larson and Marušič, 2004, i.a.):

(22) Anyone/someone/no-one asleep was woken up by the experimenters.

Why are fewer modifiers allowed by bare quantifiers than by indefinite quantifiers, then? We can relate this to the fact only PPs and unreduced relative clauses are extraposable.

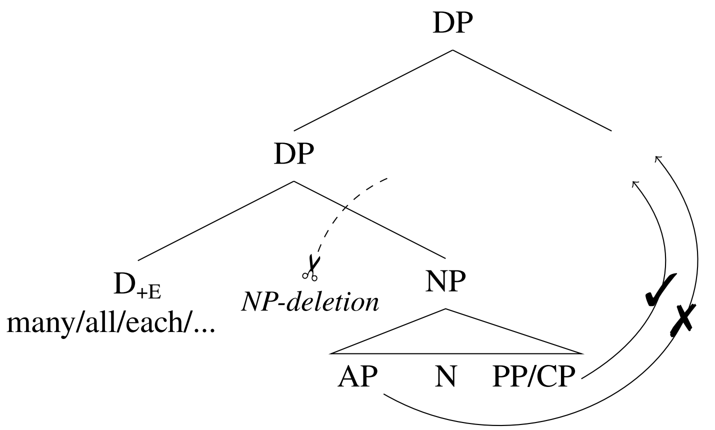
(23) a. I met the participants yesterday *(that) the experimenters examined.

b. I met the participants yesterday from the subject-pool.

c. *I met [the participants t_1] yesterday [enthusiastic/asleep]$_1$.

If bare quantifiers involved phrasal deletion of the NP-restrictor, only those modifiers that can move out of the NP before deletion could appear. I assume here – for convenience – that extraposition is rightward adjunction (see Kayne, 1994, 2005 for problems; but cf. Drummond, 2009). I also assume the quantifiers which may appear bare are lexically marked as optionally bearing the 'E(llipsis)-feature' (Merchant, 2001), which triggers deletion of their NP-complement.

(24) (NP-deletion theory)



(25) a. Some ~~$[_{\text{NP}}$ participants $t_1]$~~ [that the experimenters examined]$_1$ are here.

b. *Some ~~$[_{\text{NP}}$ t_1 participants$]$~~ [enthusiastic]$_1$ are here.

It is invoking phrasal deletion that lets us capture the fine-grained distribution of NP-modifiers with bare quantifiers. Now we extend this to quantified partitives.

3 Unifying bare quantifiers and quantified partitives with NP-deletion

Rightward adjectives and unreduced relative clauses are unavailable in quantified partitives (but PPs and unreduced relative clauses are available as shown earlier in (6)):

(26) a. Some of them *(who are) asleep are being monitored by a scientist.

b. Some of them *(that) the scientists examined showed interesting data.

That is, the NP-modifiers obey extraposability. This serves as a diagnostic for NP-deletion. Thus, I propose quantified partitives involve extraposition of the partitive-PP:

(27) Some $\cancel{[_{NP} \text{ participants } t_1\ t_2]}$ [of them]$_1$ [*(that) the scientists]$_2$ examined...

So quantified partitives are a special case of bare quantifiers, with extraposition of the partitive-PP preceding NP-deletion. This distinguishes my analysis from similar analyses such as Zamparelli 1998 and Sauerland and Yatsushiro 2004. They propose a deleted N-head, but not extraposition (their accounts also require non-constituent deletion).

The derivation above would explain why only those quantifiers which may bear the E-feature can participate in quantified partitives. As quantified partitives just are extraposed partitive PPs which are stranded after deletion, only those quantifiers which may delete their NP-complements may participate. Thus, by allowing NP-deletion in quantified partitives, we derive and unify the bareness and extraposability generalizations.

3.1 Some additional advantages to the Extraposition-plus-NP-deletion analysis

This analysis explains why partitive PPs with indefinite arguments are ungrammatical:

(28) *The researchers awoke {some/three/many} of participants.

of-NP$_{\text{indef}}$ is known as a 'pseudopartitive'. Previous analyses stipulate that indefinites do not trigger post-syntactic *of*-insertion (Jackendoff, 1977, Matthewson, 2000, 2001). The NP-deletion theory, on the other hand, reduces the contrast between quantified partitive/pseudopartitives to Selkirk's (1977) observation that pseudopartitive PPs cannot extrapose, but partitive PPs can (without committing to any particular explanation of this fact):

(29) We ate [a lot t_1] yesterday [of *(the) left-over turkey]$_1$. (adapted; Selkirk, 1977)

The NP-deletion theory also explains why the quantifier acts as if it is local to the 'downstairs' NP, despite *of the* intervening. Examples like the following show either c-selection by the quantifier of a particular number feature (Chomsky, 1965, Abney, 1987, Wiltschko, 2008), or number-conditioned allomorphy (Bresnan, 1973, Bobaljik, 2012):

(30) a. $\text{all}_{\text{SG/PL}}$ of the beer(s)

b. $\text{several}_{\text{*}\textbf{SG}\text{/PL}}$ of the beer***(s)**

(31) a. much of the beer

b. *many of the beer
(Intended: many units of the beer)

These are generally held to be highly-local, head-complement relations. The NP-deletion theory claims that there is a duplicate of the downstairs noun in a local position.

(32) a. much BEER of the beer

b. several BEERS of the beers

Despite the many advantages, there are some remaining issues.

3.2 Some potential issues with the Extraposition-plus-NP-deletion analysis

The first problem is that the analysis seems to overgenerate non-deleted forms:

(33) ??All cakes of the cakes

That is, it looks like the NP-deletion is obligatory. But as Hoeksema (1984) points out, such examples are 'redundant, and therefore rather clumsy', but not ungrammatical. He observes that the following are more acceptable:

(34) a. two cooks of those we hired last summer (Stockwell et al., 1973:120)

b. only two girls of the ten girls you recommended (Hoeksema, 1984:20)

So, NP-deletion is not *obligatory* ellipsis (cf. Elbourne, 2005 on third-person pronouns).

Another issue concerns possessive *'s*. I made the claim that all and only bare quantifiers may appear in quantified partitives. The following datapoint threatens to turn this biconditional into a one-way conditional:

(35) a. Out of the students, my classmates laughed, but John's (classmates) didn't.

b. *Out of the students, my classmates laughed, but John's of them didn't.

's may appear bare, but not with a partitive. However, *'s* is unlike bare quantifiers in that it does not allow for *any* NP-modifiers:

(36) There were many books on the shelf. I picked up {some/many/most/*John's} about the Reformation.

(37) The students all lined up. Surprisingly, {some/many/most/*John's} who were wearing colorful socks were not admonished.

So (35) either does not involve ellipsis at all, making it disanalogous to bare quantifiers altogether, or it involves deletion of a constituent which acts as an island to extraposition.

Relational nouns also pose a problem. [anonymized for review] (pers. comm.) asks why the examples in (38) are not synonymous:

(38) a. some of our friends (by hypothesis, *some ~~friends~~ of our friends*)

b. some friends of our friends

This is partially explained if the *of* which relates the nouns in *friends of John* is merely homophonous with the partitive *of* (Barker, 1998). Interestingly, unlike partitive-*of*, which allows extraposition (29), relational-*of* seems to resist it (39):

(29) We ate [a lot t_1] yesterday [of the left-over turkey]$_1$. (adapted; Selkirk, 1977)

(39) a. I met a friend of my best friend yesterday.

b. ??I met a friend t_1 yesterday [of my best friend]$_1$.

If relational-*of* PPs cannot extrapose, we block a relational/non-partitive reading of (38).

The final issue is that extraposed XPs are generally freely-ordered (Drummond, 2013):

(40) a. I tried [to give [pictures t_1 t_2] to John [that he would like]$_1$] yesterday [of his dog]$_2$.

b. I tried [to give [pictures t_1 t_2] to John [of his dog]$_1$] yesterday [that he would like]$_2$.

In quantified partitives, on the other hand, there are strict ordering restrictions. The partitive phrase must be leftmost – despite both being (by hypothesis) extraposed:

(41) a. The participants are sleeping. All ~~[$_{\text{NP}}$ participants t_1 t_2]~~ [of them]$_1$ [that we are monitoring]$_2$ are currently in R.E.M.

b. *The participants are sleeping. All ~~[$_{\text{NP}}$ participants t_1 t_2]~~ [that we are monitoring]$_1$ [of them]$_2$ are currently in R.E.M.

And multiple modifiers *are* freely-ordered with respect to one another:

(42) a. All ~~[$_{\text{NP}}$ participants t_1 t_2 t_3]~~ [of them]$_1$ [that we are monitoring]$_2$ [from the subject-pool]$_3$ are currently in R.E.M.

b. All ~~[$_{\text{NP}}$ participants t_1 t_2 t_3]~~ [of them]$_1$ [from the subject-pool]$_2$ [that we are monitoring]$_3$ are currently in R.E.M.

I currently have no explanation for why the partitive phrase must stay closest, but the above facts suggest an argument/adjunct asymmetry (see Shin, 2016 and Shah, to appear for arguments that the partitive-PP is a complement, not a specifier or adjunct).

4 Conclusion

The theory I have proposed breaks the 'quantified partitive' construction into two different, independently-motivated pieces: (i) partitive PPs can extrapose, (ii) some quantifiers can delete their NP-complements.

(43) a. Much ~~[$_{\text{NP}}$ cake]~~, much ~~[$_{\text{NP}}$ cake t_1]~~ [of the cake]$_1$

b. Many ~~[$_{\text{NP}}$ cakes]~~, many ~~[$_{\text{NP}}$ cakes t_1]~~ [of the cakes]$_1$

This theory completely unifies quantified partitives with bare quantifiers, explaining why exactly the same quantifiers appear in both, and why only extraposable NP-modifiers are available in either construction. It also avoids the overgeneration problems of a silent noun theory as well as the undergeneration problems of 'surface true' theory. In addition, it accounts for why the quantifier behaves as if it is local to the NP in the partitive PP, and reduces the unavailablility of quantified pseudopartitives to the independently observable non-extraposability of pseudopartitives. These advantages seem worth maintaining, even in the face of puzzling ordering restrictions.

References

Abney, S. P. (1987). The English Noun Phrase in its Sentential Aspect [Doctoral dissertation, Massachusetts Institute of Technology].

Bach, E., & Cooper, R. (1978). The NP-S Analysis of Relative Clauses and Compositional Semantics. Linguistics and Philosophy, 2(1), 145–150.

Barker, C. (1998). Partitives, Double Genitives and Anti-Uniqueness. Natural Language & Linguistic Theory, 16(4), 679–717.

Barwise, J., & Cooper, R. (1981). Generalized Quantifiers and Natural Language. Linguistics and Philosophy, 4(2), 159–219.

Bobaljik, J. D. (2012). Universals in Comparative Morphology: Suppletion, Superlatives, and the Structure of Words. The MIT Press.

Bresnan, J. (1973). Syntax of the Comparative Clause Construction in English. Linguistic Inquiry, 4(3), 275–343.

Chomsky, N. (1965). Aspects of the Theory of Syntax. The MIT Press.

Dobrovie-Sorin, C., & Giurgea, I. (2021). Majority Quantification and Quantity Superlatives: A Crosslinguistic Analysis of Most. Oxford University Press.

Drummond, A. (2009). The Unity of Extraposition and the A/Ā Distinction. Proceedings from the Annual Meeting of the Chicago Linguistic Society, 45, 43–56.

Drummond, A. (2013). The case for extraposition as movement [Handout from a talk given at University of Massachusetts Amherst, accessed from https://adrummond.net/umasstalk2.pdf].

Elbourne, P. D. (2005). Situations and Individuals. MIT Press.

Gagnon, M. R. (2013). Anaphors and the Missing Link [University of Maryland, College Park dissertation].

Hoeksema, J. (1984). Partitives [Unpublished manuscript, R.U. Groningen.].

Jackendoff, R. S. (1968). Quantifiers in English. Foundations of Language, 4(4), 422–442.

Jackendoff, R. S. (1971). Gapping and Related Rules. Linguistic Inquiry, 2(1), 21–35.

Jackendoff, R. S. (1977). X-Bar Syntax: A Study of Phrase Structure. The MIT Press.

Kayne, R. S. (1994). The Antisymmetry of Syntax. MIT Press.

Kayne, R. S. (2005). Movement and Silence. Oxford University Press.

Kayne, R. S. (2007). On the Syntax of Quantity in English. In J. Bayer, T. Bhattacharya, & M. H. Babu (Eds.), Linguistik Aktuell/Linguistics Today (pp. 73–105, Vol. 102). John Benjamins Publishing Company.

Kayne, R. S. (2017). English One and Ones as Complex Determiners. In G. Sengupta, S. Sircar, M. G. Raman, & R. Balusu (Eds.), Perspectives on the Architecture and Acquisition of Syntax (pp. 77–114). Springer Singapore.

Kayne, R. S. (2025). Antihomophony and English of [Available from https://ling.auf.net/lingbuzz/009062].

Larson, R. K., & Marušič, F. (2004). On Indefinite Pronoun Structures with APs: Reply to Kishimoto. Linguistic Inquiry, 35(2), 268–287.

Lobeck, A. (1995). Ellipsis: functional heads, licensing, and identification. Oxford University Press.

Matthewson, L. (2000). Quantification and (the Absence of) Cross-Linguistic Variation. University of Massachusetts Occassional Papers in Linguistics, 23, 101–135.

Matthewson, L. (2001). Quantification and the Nature of Crosslinguistic Variation. Natural Language Semantics, 9(2), 145–189.

Merchant, J. (2001). The Syntax of Silence: Sluicing, Islands, and the Theory of Ellipsis. Oxford University Press.

Merchant, J. (2023). Nominal ellipsis, n/N heads, and lexical nominalization [Handout, Colloquium on Generative Grammar 32; 26 April 2023, University of the Basque Country, Vitoria/Gasteiz. Available from https://home.uchicago.edu/merchant/pubs/CGG32.NPE.Merchant.hd.2up.pdf].

Sauerland, U., & Yatsushiro, K. (2004). A Silent Noun in Partitives. Proceedings of NELS 34, Keir Moulton and Matthew Wolf (eds.), 101–112.

Selkirk, E. O. (1977). Some Remarks on Noun Phrase Structure. In P. Culicover, T. Wasow, & A. Akmajian (Eds.), Formal Syntax (pp. 285–317). Academic Press.

Shah, M. P. (to appear). Recursion in NP: pseudopartitive measures require complementation, not specification. [Available from https://ling.auf.net/lingbuzz/009325]. Proceedings of WCCFL43.

Shin, K. Y. (2016). Quantifiers in English Partitives. Natural Language & Linguistic Theory, 34(4), 1493–1508.

Stockwell, R. P., Schachter, P., & Partee, B. H. (1973). The Major Syntactic Structures of English. Holt, Rinehart & Winston.

Wiltschko, M. (2008). The Syntax of Non-Inflectional Plural Marking. Natural Language & Linguistic Theory, 26(3), 639–694.

Zamparelli, R. (1998). A Theory of Kinds, Partitives and of/z Possessives. In A. Alexiadou & C. Wilder (Eds.), Possessors, Predicates and Movement in the Determiner Phrase (pp. 259–302, Vol. 22). John Benjamins Publishing Company.